

양자컴퓨터 이후의 보안: 양자내성암호와 표준화

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최형민

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- 2024 한국 국가암호공모전 대상, 2020 iDASH 우승 등



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Why Post-Quantum Cryptography?





Why Post-Quantum Cryptography?

현대암호

Smart ID/MS, Smart Security
SOFTFORUM

인증서 위치

하드디스크 이동식디스크 저장토큰 보안토큰 휴대폰

구분	사용자	만료일	발급자
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인증서 찾기 인증서 암호는 대소문자를 구분합니다.

인증서 보기 인증서 암호

인증서 삭제 ! 인증서 선택 후 암호를 입력하세요.

확인 취소

https://eprint.iacr.org

eprint.iacr.org

이 연결은 안전합니다.

쿠키 및 사이트 데이터

사이트 설정

주의 요함 golang.site

golang.site

이 사이트는 보안 연결(HTTPS)이 사용되지 않았습니다.
이 사이트에 입력하는 비밀번호나 신용카드 번호 등의 정보가 공격자에 의해 도용될 수 있습니다. [자세히 알아보기](#)





Why Post-Quantum Cryptography?

현대암호

인증서

일반

자세히

인증 경로

표시(S):

<전체>

필드	값
버전	V3
일련 번호	0f14
서명 알고리즘	sha256RSA
서명 해시 알고리즘	sha256
발급자	
유효 기간(시작)	2020년 7월 22일 수요일 오...
유효 기간(끝)	2024년 10월 8일 화요일 오...
주체	RSA TLS CA 01, ...

sha256RSA

인증서 필드

WE1

인증서

버전

일련번호

인증서 서명 알고리즘

발행기관

유효성

시작:

필드 값

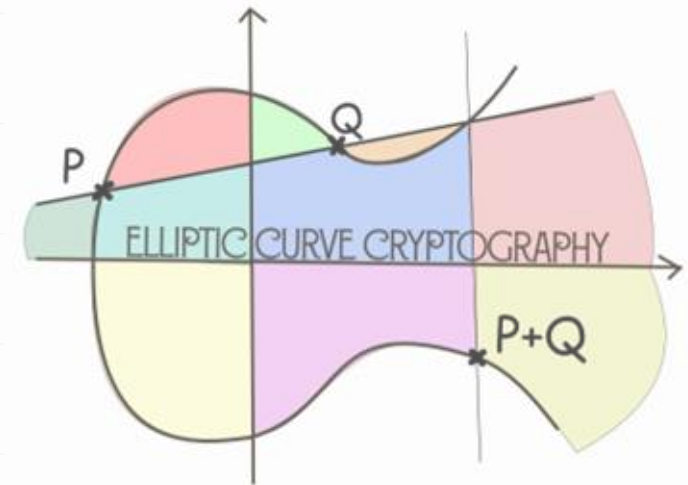
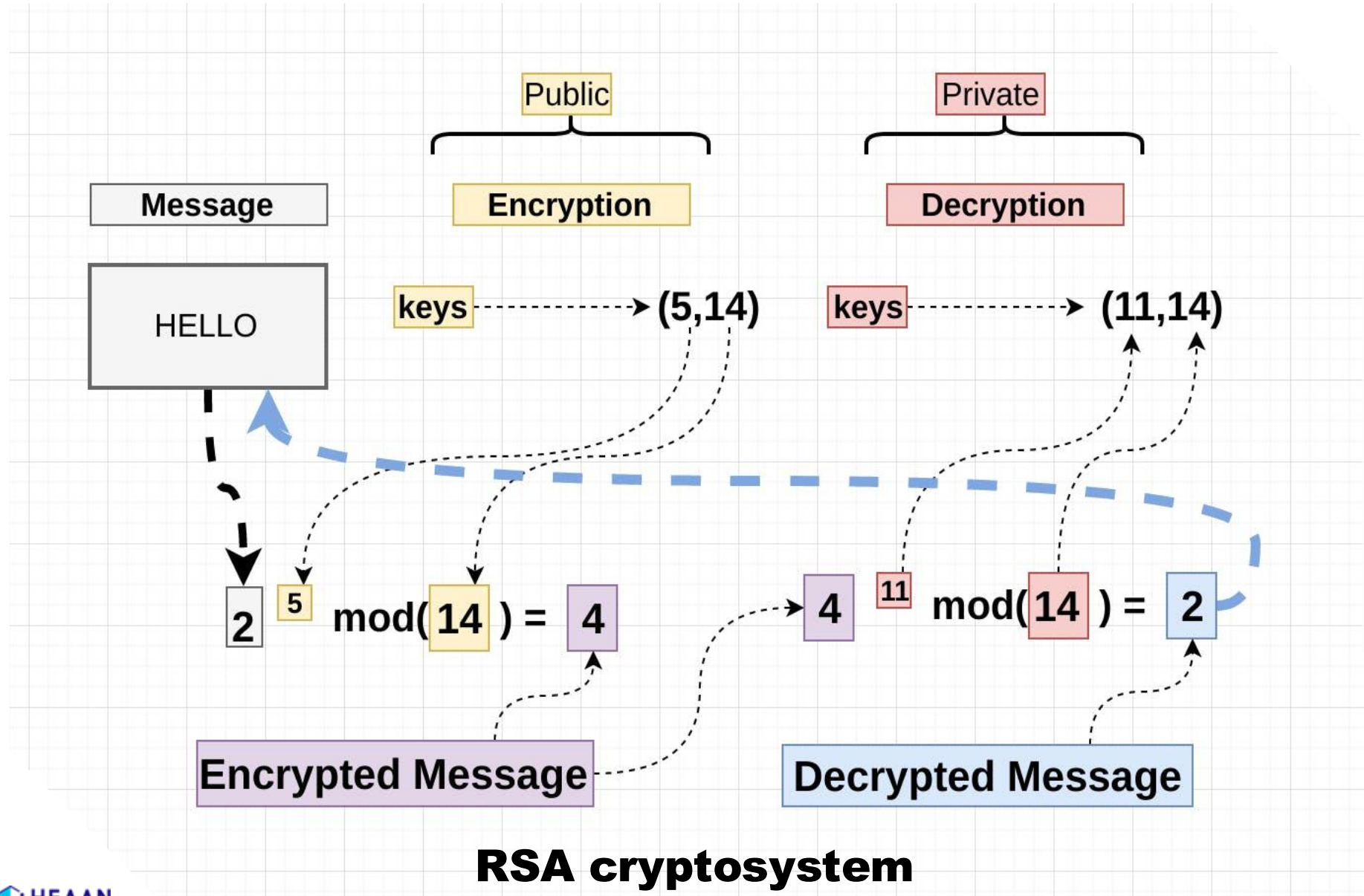
SHA-384를 포함한 X9.62 ECDSA 서명

내보내기(X)...



Why Post-Quantum Cryptography?

현대암호



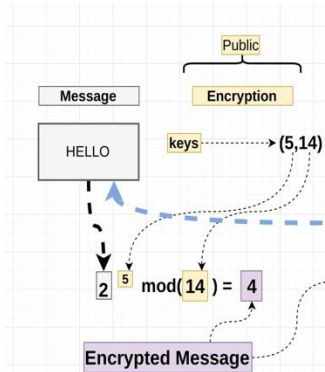
EC cryptosystem



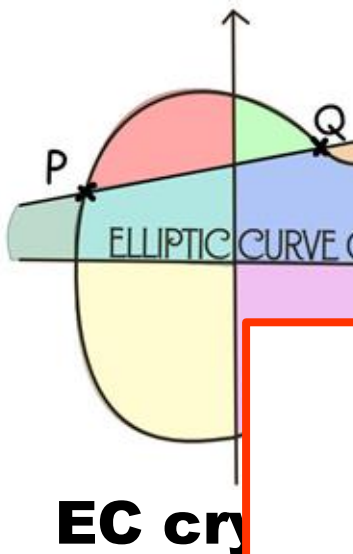
Why Post-Quantum Cryptography

“Pre-Quantum” 혹은 “Classical”

암호 시스템

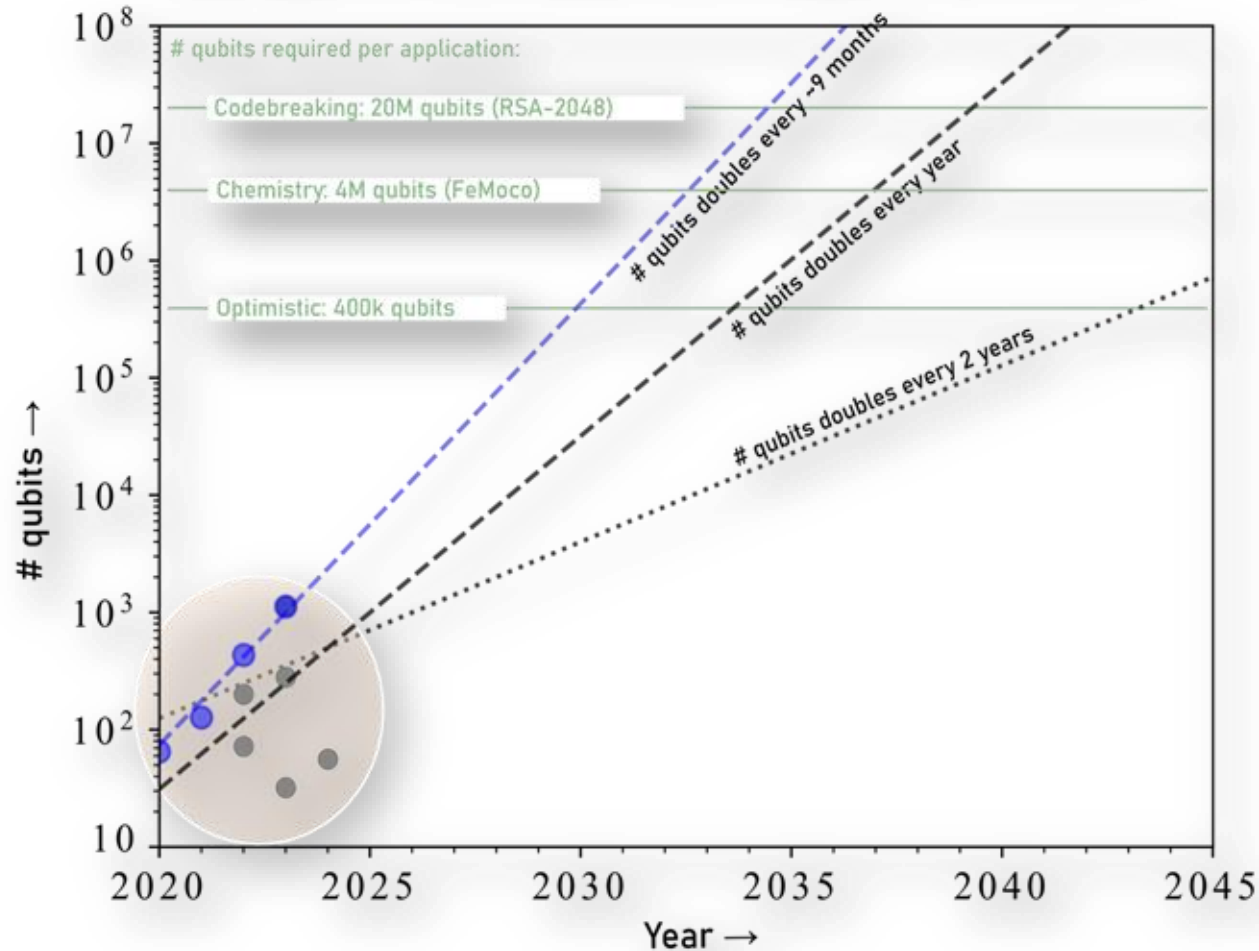


RSA crypt



EC cry

Qubit growth estimates, according to Moore's Law



RSA2048: billions of years vs. several hours in 4,099 **logical**-qubit quantum computer.
VERY recent estimation: a week in 1M **noisy**-qubit QC.

ard Problems)

horiza
의 인 해

, $e, n \in$

gorithm
Computer

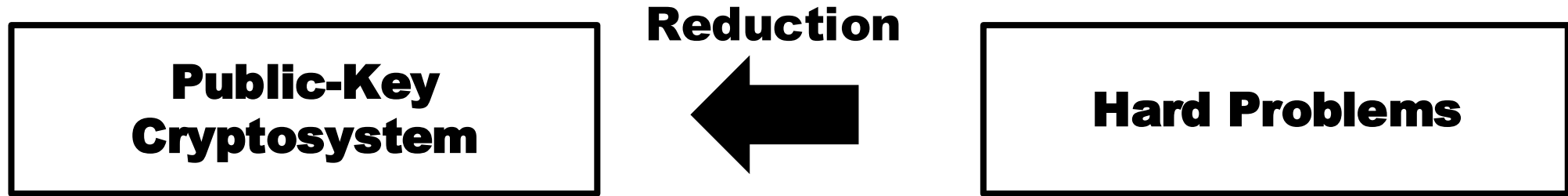
problem:
and P,



Why Post-Quantum Cryptography?

Post-Quantum Cryptography 양자내성암호

- 현대암호의 안전성은 수학적 난제를 기반으로 보장.





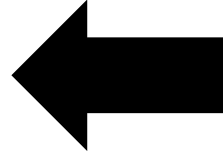
Why Post-Quantum Cryptography?

Post-Quantum Cryptography 양자내성암호

- 양자내성암호는 양자컴퓨터로도 풀기 어려운 수학적 난제들을 사용.

**Post-Quantum
Public-Key
Cryptosystem**

Reduction



**Hard problems
(even) against
Quantum Algorithms**

- Lattice-based cryptography

- Code-based cryptography

•
•
•

- Shortest/Closest Vector Problem (SVP/CVP)

≈ Learning With Errors (LWE)

≈ Learning With Rounding (LWR)

- Syndrome Decoding Problem (SDP)

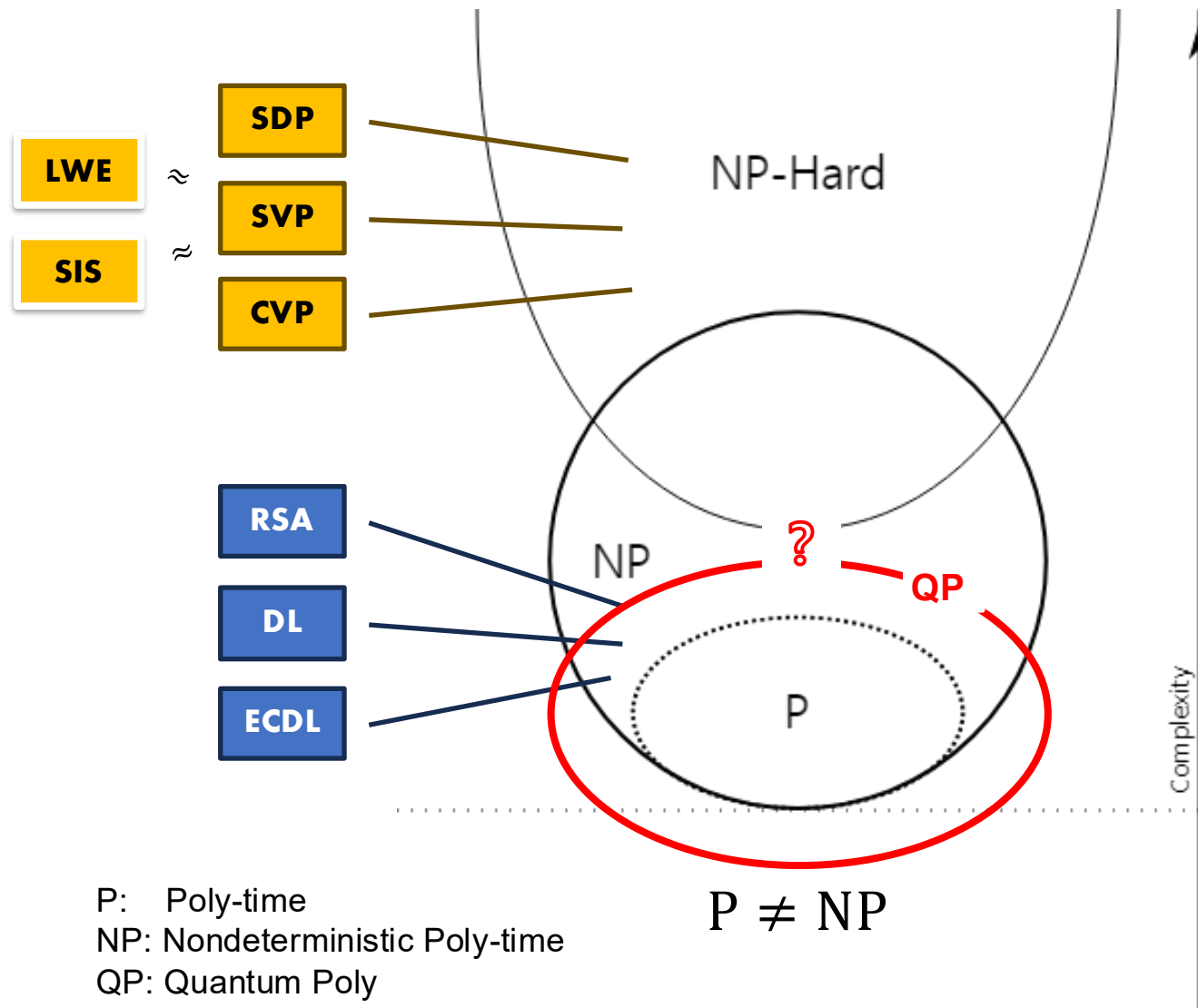
•
•
•

NP-hard!*



Why Post-Quantum Cryptography?

Post-Quantum Cryptography 양자내성암호



NIST PQC Competition and KpqC Competition

NIST

Information Technology Laboratory

COMPUTER SECURITY RESOURCE CENTER

PROJECTS

Post-Quantum Cryptography PQC



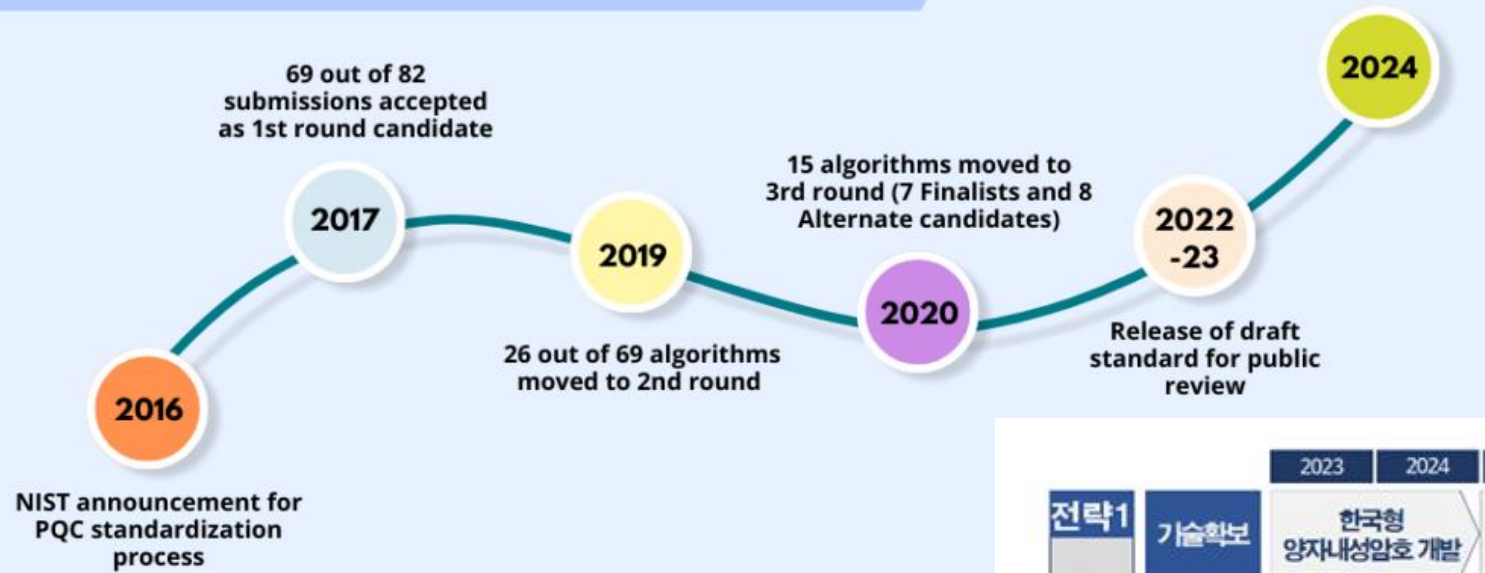
양자내성암호연구단



Why Post-Quantum Cryptography?

Global Movement

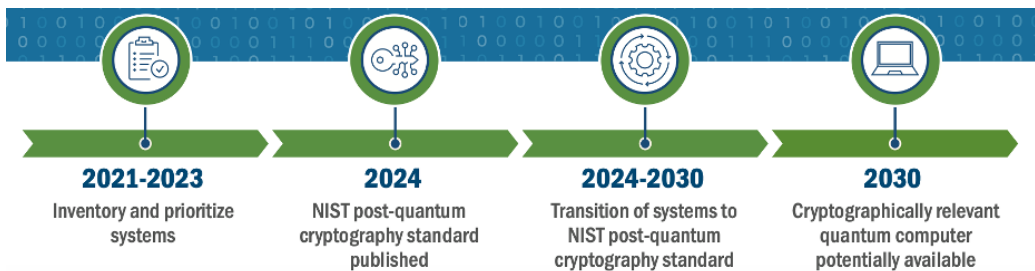
NIST PQC Standardization Milestones



ETSI White Paper No. 8

Quantum Safe Cryptography and Security

An introduction, benefits, enablers and challenges





NIST PQC Competition

- 2016년, 미국 국립표준기술연구소(NIST)에서 양자내성암호의 표준을 공모
 - 공개 키 암호(PKE)와 키 캡슐화(KEM) 분야
 - 전자서명(Digital Signature) 분야

2017 Round 1	PKE/KEM	Signatures	Overall
Lattice-based	21	5	26
Code-based	17	2	19
Multivariate	2	7	9
Hash-based	0	2	2
Isogeny-based	1	0	1
Other	4	3	7
Total	45	19	64



NIST PQC Competition

- 2016년, 미국 국립표준기술연구소(NIST)에서 양자내성암호의 표준을 공모
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 - 전자서명(Digital Signature) 분야

2019 Round 2	PKE/KEM	Signatures	Overall
Lattice-based	9	3	12
Code-based	7	0	7
Multivariate	0	4	4
Hash-based	0	1	1
Isogeny-based	1	0	1
Other	0	1	1
Total	17	9	26



- 2016년, 미국 국립표준기술연구소(NIST)에서 양자내성암호의 표준을 공모
 - 공개 키 암호(PKE)와 키 캡슐화(KEM) 분야
 - 전자서명(Digital Signature) 분야

* Finalists

* Alternate candidates

2020 Round 3	PKE/KEM	Signatures	Overall
Lattice-based	3+2	2	5+2
Code-based	1+2	0	1+2
Multivariate	0	1+1	1+1
Hash-based	0	1	1
Isogeny-based	1	0	1
Other	0	1	1
Total	4+5	3+3	7+8



NIST PQC Competition

- 2016년, 미국 국립표준기술연구소(NIST)에서 양자내성암호의 표준을 공모
 - 공개 키 암호(PKE)와 키 캡슐화(KEM) 분야
 - 전자서명(Digital Signature) 분야

2022 Selected	PKE/KEM	Signatures	Overall
Lattice-based	CRYSTALS-Kyber	CRYSTALS-Dilithium FALCON	3
Hash-based		SPHINCS+	1

2025 Selected	PKE/KEM	Signatures	Overall
Code-based	HQC		1

Overall	2	3	5
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KpqC Competition

- 2021년, 국립보안기술연구소 (NSRI)에서 한국형 양자내성암호 확보를 목표로 공모전 시작
 - 공개 키 암호(PKE)와 키 캡슐화(KEM) 분야
 - 전자서명(Digital Signature) 분야

2022 Round 1	PKE/KEM	Signatures	Overall
Lattice-based	3	5	8
Code-based	4	1	5
Multivariate	0	1	1
Hash-based	0	1	1
Symmetric-based	0	1	1
Total	7	9	16



KpqC Competition

- 2021년, 국립보안기술연구소 (NSRI)에서 한국형 양자내성암호 확보를 목표로 공모전 시작
 - 공개 키 암호(PKE)와 키 캡슐화(KEM) 분야
 - 전자서명(Digital Signature) 분야

2023 Round 2	PKE/KEM	Signatures	Overall
Lattice-based	2	2	4
Code-based	2	0	2
Multivariate	0	1	1
Hash-based	0	0	0
Symmetric-based	0	1	1
Total	4	4	8



KpqC Competition

- 2021년, 국립보안기술연구소 (NSRI)에서 한국형 양자내성암호 확보를 목표로 공모전 시작
 - 공개 키 암호(PKE)와 키 캡슐화(KEM) 분야
 - 전자서명(Digital Signature) 분야

2025 Selected	PKE/KEM	Signatures	Overall
Lattice-based	SMAUG-T NTRU+	HAETAE	3
Symmetric-based		AlMer	1
Overall	2	2	4





SMAUG-T

HEΛΛN
CRYPTO LAB



Defense Counterintelligence
Command

NIST and KpqC Standards



HAETAE

HEΛΛN
CRYPTO LAB



SMAUG-T

HEΛΛN
CRYPTO LAB



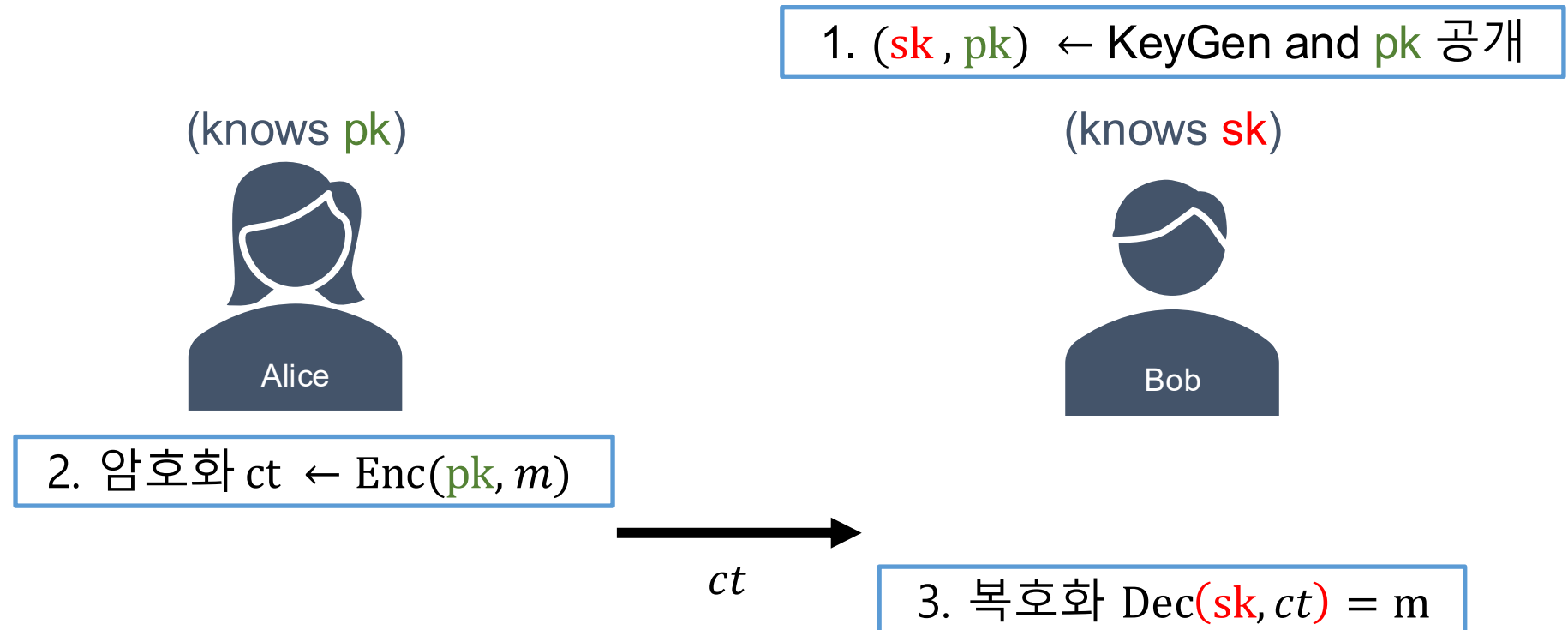
Defense Counterintelligence
Command

Kyber and SMAUG-T



공개 키 암호화 방식 (PKE)

- 공개 키 암호화 방식 Public Key Encryption (PKE)
 - Alice → Bob 메시지 전송하되, **Confidentiality**가 보장되기를 바람.
 - 즉, Bob 이외에는 메시지를 볼 수 없기를 바람.





키 캡슐화 메커니즘 (KEM)

- 키 캡슐화 메커니즘 Key Encapsulation Mechanism (KEM)
 - Alice가 Bob과 대칭키암호를 사용하고 싶을 때, 그에 앞서 공통의 키를 공유

1. $(sk, pk) \leftarrow \text{KeyGen}$ and pk 공개

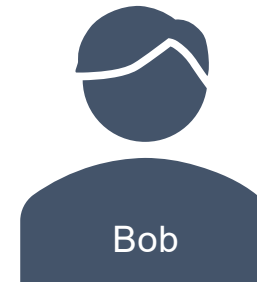
(knows sk)



3. 디캡슐화

$K \leftarrow \text{Decap}(sk, ct)$

(knows pk)



2. 캡슐화

$K, ct \leftarrow \text{Encap}(pk; r)$

ct

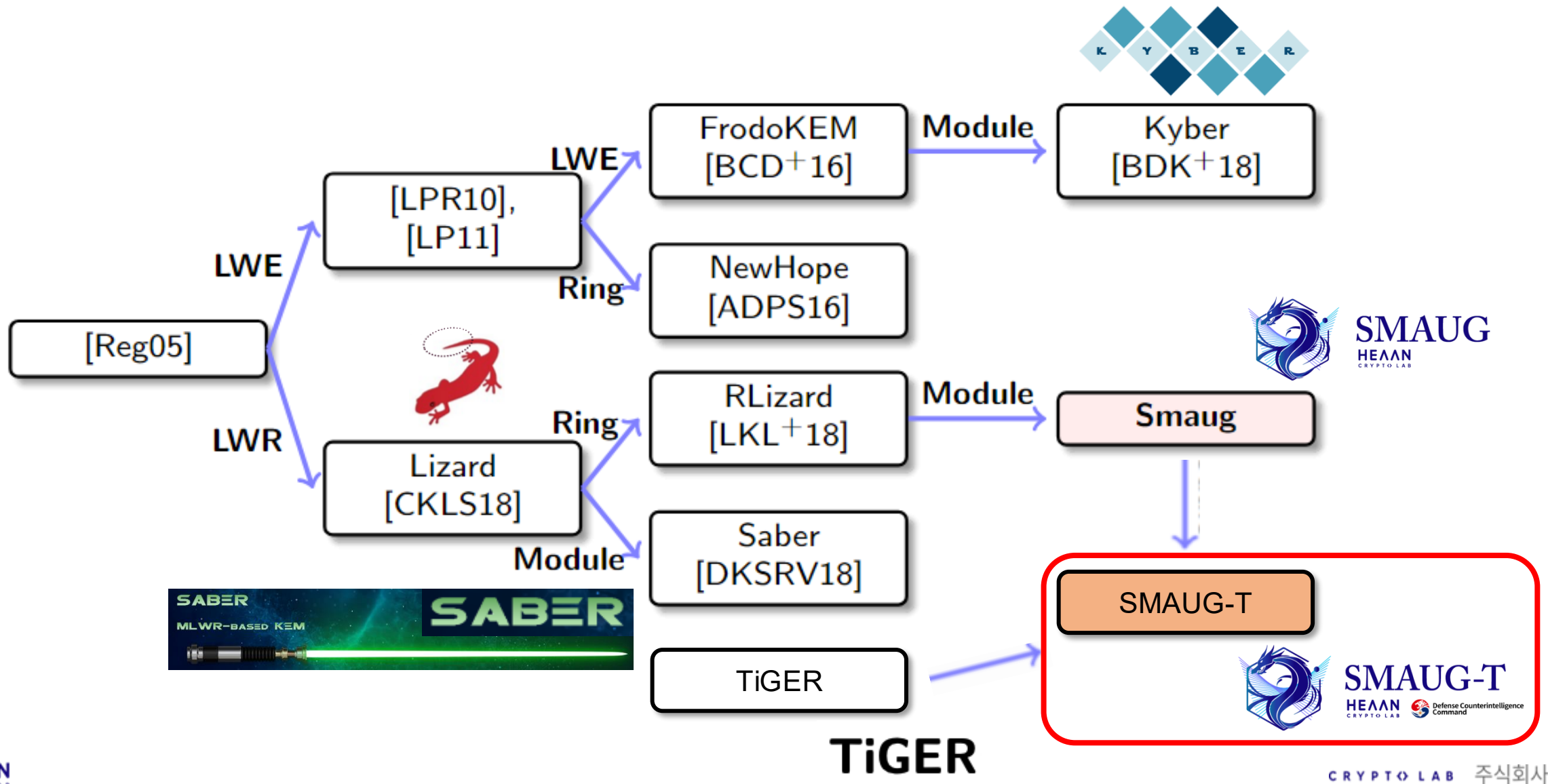
ct_{SKE}

4. 대칭키암호를 이용한 보안 통신

$ct_{SKE} \leftarrow \text{Enc}(K, m) \quad \& \quad m \leftarrow \text{Dec}(K, ct_{SKE})$



THE History





- **Learning With Errors (LWE)**

- Gaussian noise is added:

$$\left(\begin{matrix} n \\ m \end{matrix} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \mathbf{A}, \begin{matrix} \mathbf{b} = \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \mathbf{A} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \mathbf{s} + \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \mathbf{e} \end{matrix} \right) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$$

- **Learning With Rounding (LWR)**

- Noise comes from rounding:

$$\left(\begin{matrix} n \\ m \end{matrix} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \mathbf{A}, \begin{matrix} \mathbf{b} = \left\lfloor \frac{p}{q} \cdot \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \mathbf{A} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \mathbf{s} \right\rfloor \end{matrix} \right) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_p^m$$

- **Ring variant (RLWE)**

$$\left(\textcircled{a(x)}, \textcircled{b(x)} = a(x) \cdot \textcircled{s(x)} + \textcircled{e(x)} \right) \in R_q \times R_q$$

- **Ring variant (RLWR)**

$$\left(\textcircled{a(x)}, \textcircled{b(x)} = \left\lfloor \frac{p}{q} \cdot a(x) \cdot \textcircled{s(x)} \right\rfloor \right) \in R_q \times R_p$$

n : 2의 거듭제곱
 $R = \mathbb{Z}[x]/(x^n + 1)$: 다항식 환 Ring



- **Module LWE (MLWE)**

- Gaussian noise is added:

$$(a_{0,0}(x) \in R_q, \begin{matrix} n \\ \text{---} \\ (m) \left[\begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \right] \mathbf{A} \end{matrix}, \begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \mathbf{b} = \begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \mathbf{A} \begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \mathbf{s} + \begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \mathbf{e}) \in R_q^{m \times n} \times R_q^m$$

- **Module LWR (MLWR)**

- Noise comes from rounding:

$$(\begin{matrix} n \\ \text{---} \\ (m) \left[\begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \right] \mathbf{A} \end{matrix}, \begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \mathbf{b} = \left\lfloor \frac{p}{q} \cdot \begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \mathbf{A} \begin{matrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{matrix} \mathbf{s} \right\rfloor) \in R_q^{m \times n} \times R_p^m$$

- **Ring variant (RLWE)**

$$(a(x), b(x) = a(x) \cdot s(x) + e(x)) \in R_q \times R_q$$

- **Ring variant (RLWR)**

$$(a(x), b(x) = \left\lfloor \frac{p}{q} \cdot a(x) \cdot s(x) \right\rfloor) \in R_q \times R_p$$

n : power of two integer
 $R = \mathbb{Z}[x]/(x^n + 1)$



$$n = 256$$

$$R = \mathbb{Z}[x]/(x^n + 1)$$

• Kyber.PKE

• MLWE-based Public Key

- secret (s) & noise (e): CBD

pk: $\left(\begin{matrix} k \\ k \end{matrix} \left[\begin{matrix} \text{A} \end{matrix} \right], \text{b} = \begin{matrix} \text{A} \end{matrix} \text{s} + \begin{matrix} \text{e} \end{matrix} \right), \text{sk}: \text{s}$

Each elt. is polynomial in $\mathcal{R}_q$

$$\text{A} \in R_q^{k \times k}, \text{b} \in R_q^k \text{ for } k = 2, 3, 4$$

• MLWE-based Encryption

- random (r) & noise (e): CBD

$$\left(\begin{matrix} \text{r} \end{matrix} \begin{matrix} \text{A} \end{matrix} + \begin{matrix} \text{e} \end{matrix}, \begin{matrix} \text{r} \end{matrix} \begin{matrix} \text{b} \end{matrix} + \begin{matrix} \Delta \text{M} \end{matrix} + \begin{matrix} \text{e} \end{matrix} \right)$$

$$\text{r} \in R_q^k$$

• SMAUG-T.PKE

• MLWE-based Public Key

- secret (s): sparse ternary, noise (e): dGaussian

• MLWR-based Encryption

- random (r): sparse CBD, no noise sampling

$$\left(\left[\frac{p}{q} \cdot \begin{matrix} \text{r} \end{matrix} \begin{matrix} \text{A} \end{matrix} \right], \left[\frac{p'}{q} \cdot \begin{matrix} \text{r} \end{matrix} \begin{matrix} \text{b} \end{matrix} + \frac{p'}{q} \cdot \begin{matrix} \Delta \text{M} \end{matrix} \right] \right)$$



$$n = 256$$

$$R = \mathbb{Z}[x]/(x^n + 1)$$

- Kyber.PKE Dec

- $$-c_1 s + c_2 \approx \Delta M + \tilde{e} \pmod{q}$$

$$\left(\underbrace{\left(\begin{array}{|c|} \hline r \\ \hline \end{array} \begin{array}{|c|c|} \hline & A \\ \hline \end{array} + \begin{array}{|c|} \hline e_1 \\ \hline \end{array} \right)}_{c_1}, \underbrace{\left(\begin{array}{|c|} \hline r \\ \hline \end{array} \begin{array}{|c|} \hline b \\ \hline \end{array} + \begin{array}{|c|} \hline \Delta M \\ \hline \end{array} + \begin{array}{|c|} \hline e_2 \\ \hline \end{array} \right)}_{c_2} \right)$$

$$-c_1 s + c_2 \approx - \begin{array}{|c|} \hline r \\ \hline \end{array} \begin{array}{|c|c|} \hline & A \\ \hline \end{array} \begin{array}{|c|} \hline s \\ \hline \end{array} - \begin{array}{|c|} \hline e_1 \\ \hline \end{array} \begin{array}{|c|} \hline s \\ \hline \end{array} + \begin{array}{|c|} \hline r \\ \hline \end{array} \left(\begin{array}{|c|c|} \hline & A \\ \hline \end{array} \begin{array}{|c|} \hline s \\ \hline \end{array} + \begin{array}{|c|} \hline e \\ \hline \end{array} \right) + \begin{array}{|c|} \hline \Delta M \\ \hline \end{array} + \begin{array}{|c|} \hline e_2 \\ \hline \end{array}$$

= 만약 $\tilde{e} > \frac{\Delta}{2}$ 이면 M 복구 X. → Decryption Fails!

→ Decryption Failure Rate은 아주 $O(2^{-\lambda})$ 정도로 설정

- SMAUG-T.PKE Dec

- $$-c_1 s \cdot \frac{q}{p} + c_2 \cdot \frac{q}{p'} \approx \Delta M + \tilde{e} \pmod{q}$$

Dilithium and HAETAE



HAETAE
HEΛΛN
CRYPTO LAB

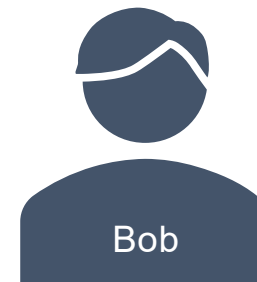


- 전자서명 Digital Signature

- Alice → Bob 메시지 전송하되, **무결성**이 보장되기를 바람.
- 즉, 메시지가 위변조되지 않기를 바람 (**위조불가능성**).
- 즉, Bob이 받은 메시지가 Alice가 보낸 메시지임을 확신할 수 있기를 바람.



문서에 서명



Alice가 보냈음을 검증



• 전자서명 Digital Signature

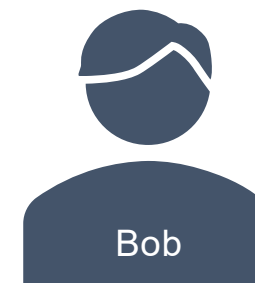
- Alice → Bob 메시지 전송하되, **무결성**이 보장되기를 바람.
- 즉, 메시지가 위변조되지 않기를 바람 (**위조불가능성**).
- 즉, Bob이 받은 메시지가 Alice가 보낸 메시지임을 확신할 수 있기를 바람.

1. (**sk**, **pk**) ← KeyGen and **pk** 공개

(knows **sk**)



(knows **pk**)



2. 서명 $\sigma \leftarrow \text{Sign}(\mathbf{sk}, m)$

→
(m, σ)

3. 검증 $\text{Verify}(\mathbf{pk}, m, \sigma)$
= accept (or reject)



“Fiat-Shamir with Abort” Paradigm

<KeyGen>

- **Secret key** : $S \in R_q^{m \times k}$
- **Public key** : $A \in R_q^{n \times m}, T \in R_q^{n \times k} (= AS \bmod q)$

<Signing>

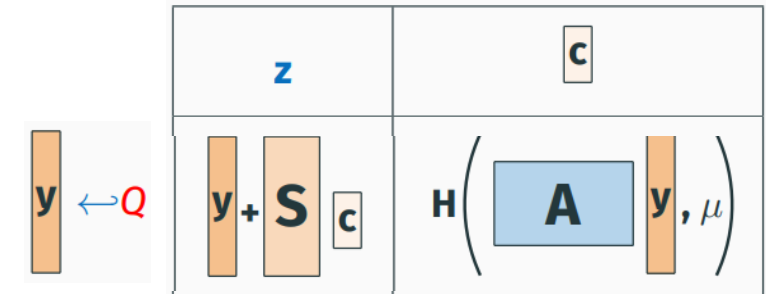
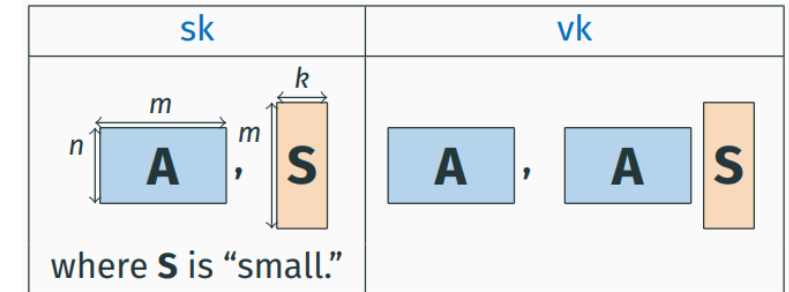
- pick y randomly
- compute $c = H(Ay \bmod q, \mu) \in R_q$
- signature: $(z = y + Sc, c) \in R_q^m \times R_q$ in certain criteria via **rejection sampling**.

<Verifying>

- Check if z small &

$$c = H(Az - Tc \bmod q, \mu)$$

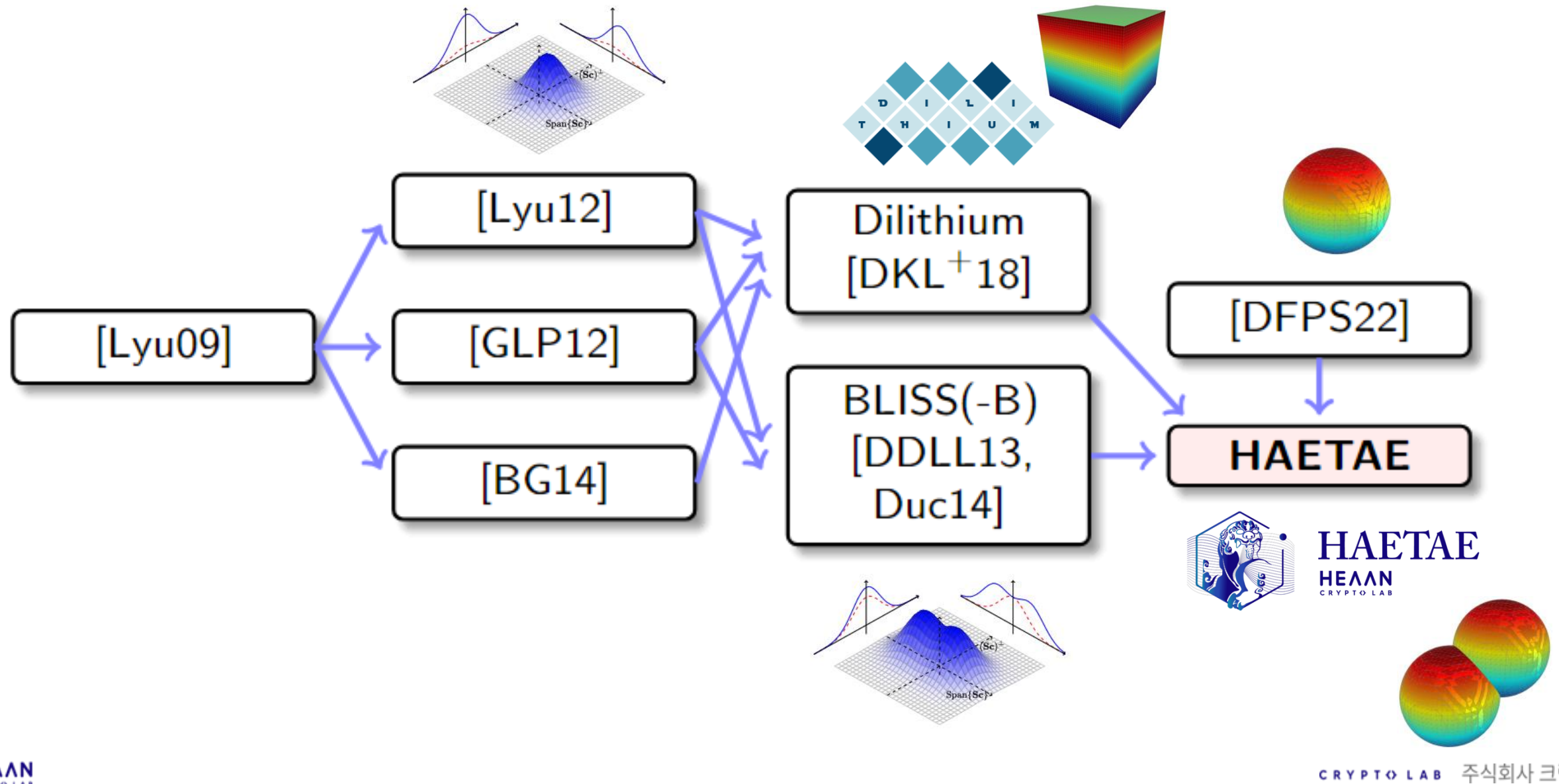
$$\begin{bmatrix} A \\ z \end{bmatrix} - \begin{bmatrix} T \\ c \end{bmatrix} = \begin{bmatrix} A \\ y + S \end{bmatrix} \begin{bmatrix} c \end{bmatrix} - \begin{bmatrix} A \\ S \end{bmatrix} \begin{bmatrix} c \end{bmatrix} = \begin{bmatrix} A \\ y \end{bmatrix}$$



$$z = y + S \begin{bmatrix} c \end{bmatrix} \text{ depends on } S.$$



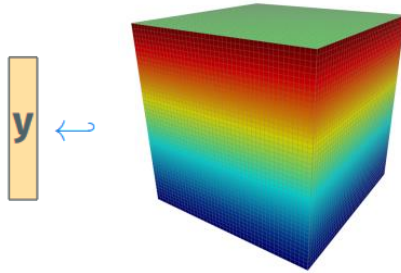
THE History



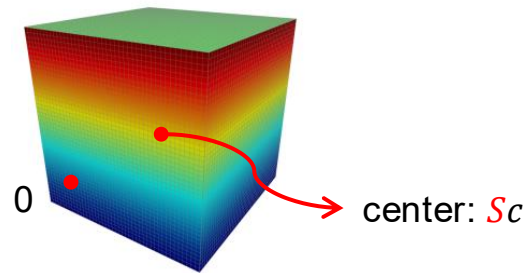


- Dilithium

- Randomness y

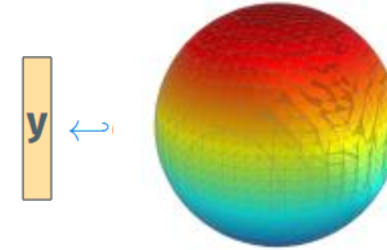


- $z = y + Sc$

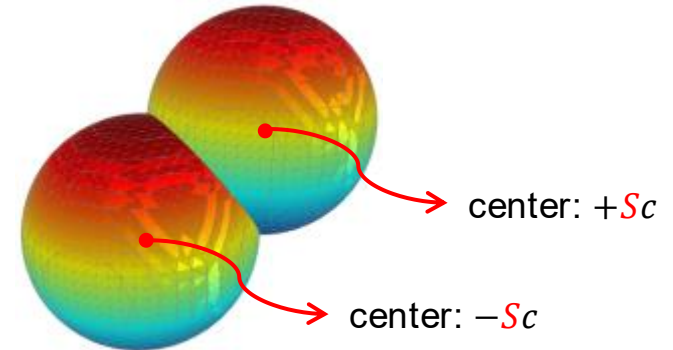


- HAETAE

- Randomness y



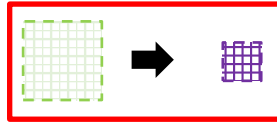
- $z = y + (-1)^b \cdot Sc$



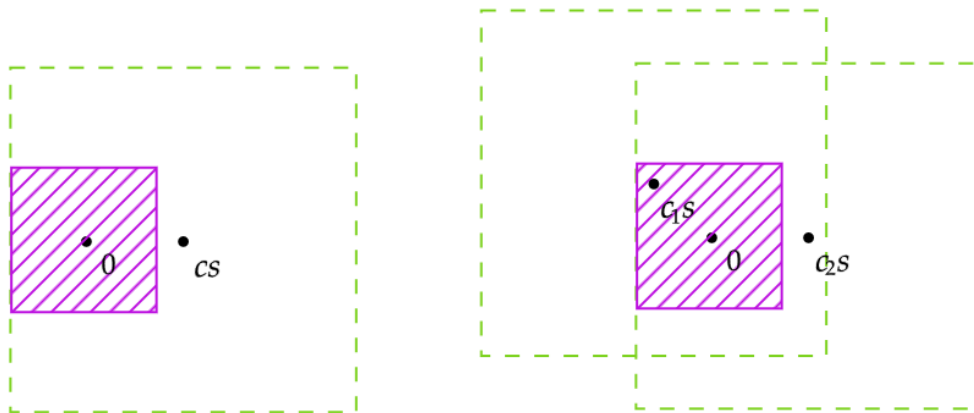


- Dilithium

- Rejection sampling



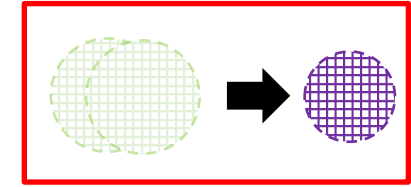
- From: **hypercube** centered at Sc
 - To: **smaller hypercube** centered at 0



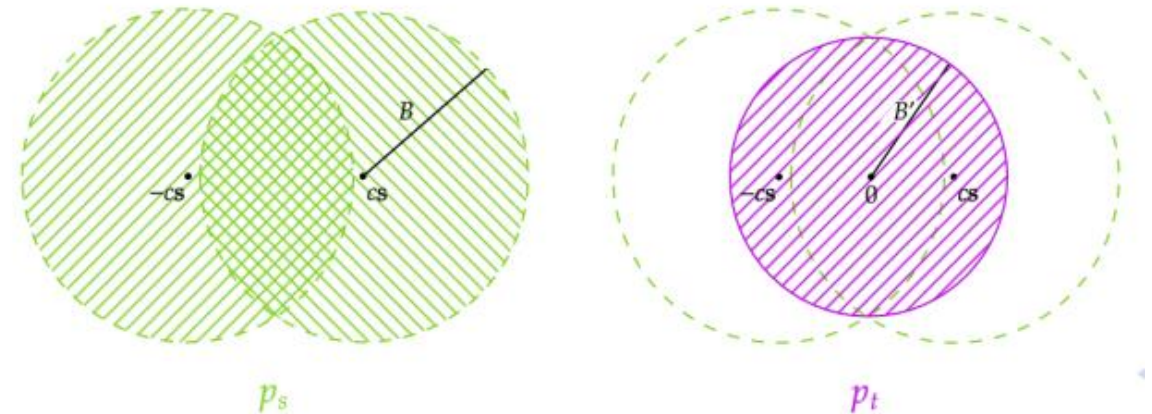
- By checking $\|z\|_\infty < t$, accept or reject

- HAETAE

- Rejection sampling



- From: **hyperball** centered at $\pm Sc$
 - To: **smaller hyperball** centered at 0

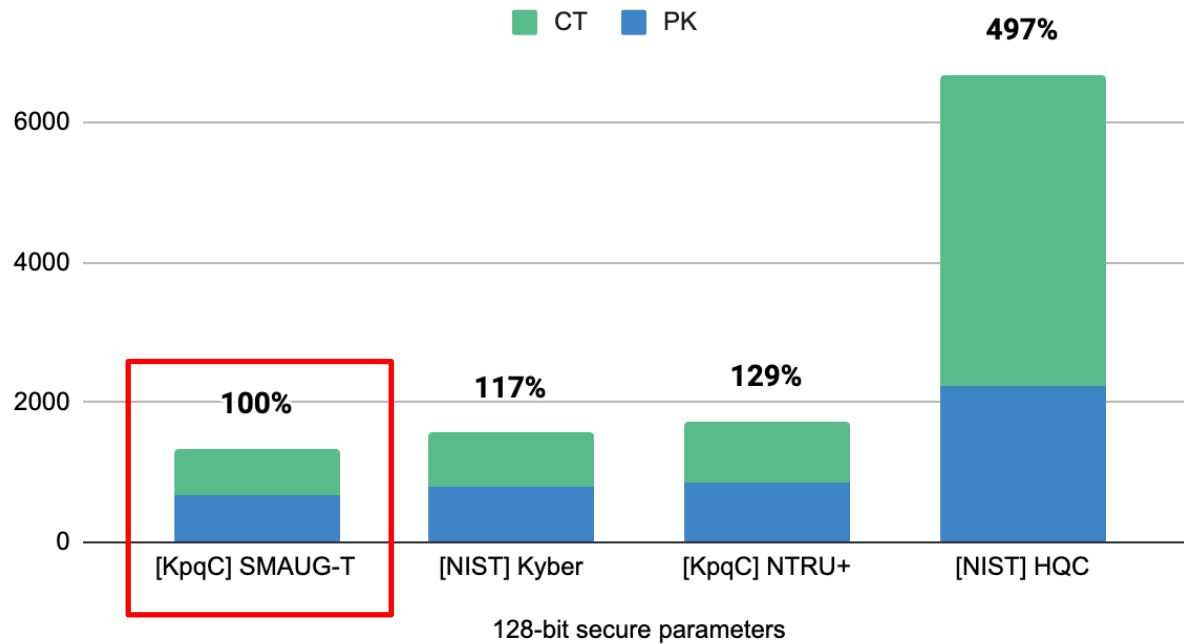


- By checking $\|z\|_2 < B'$ and some more, accept or reject



KEM

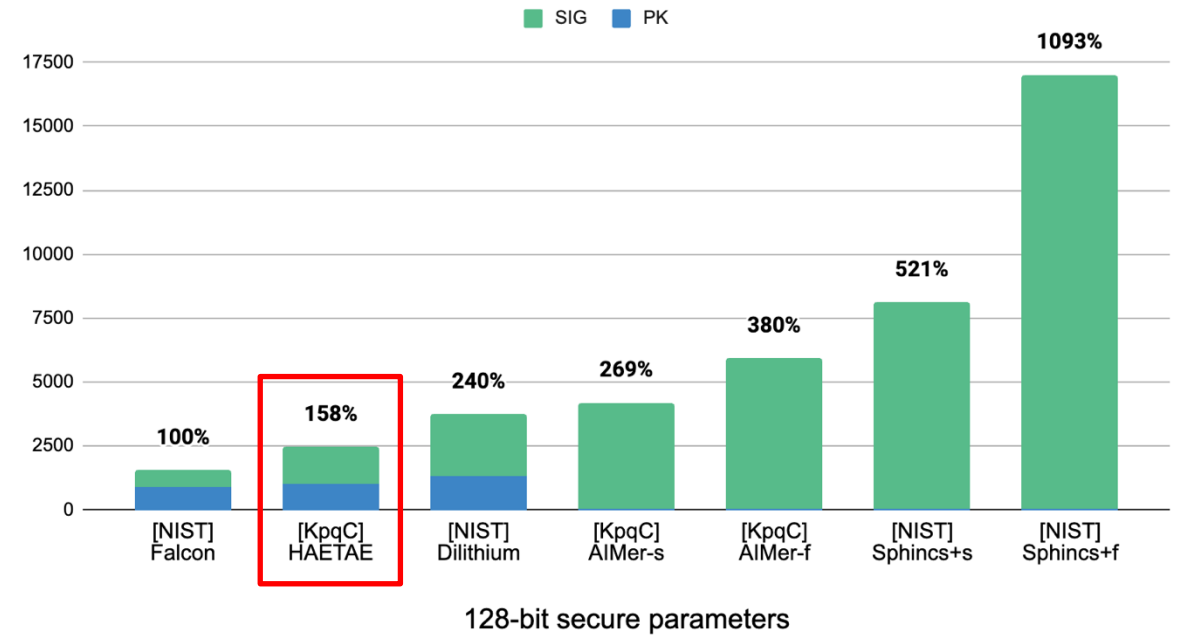
Size in bytes



- Concretely-proven security
- Small sizes with good performance

Digital Signatures

Size in bytes



NIST

Information Technology Laboratory

COMPUTER SECURITY RESOURCE CENTER

PROJECTS

Post-Quantum Cryptography PQC



양자내성암호연구단

From Competition to Standard



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산업표준심의회
2023년 4월 3일 개정



- **Migration to PQC**

- When? ASAP, Harvest Now, Decrypt Later (HNDL) attacks!



x: time that products and data must remain **secure**

y: time it takes to **migrate** to post-quantum cryptography

z: time it takes until cryptographically-relevant **quantum computers will be available**



Future Plan

- KS 대한민국 표준
- ISO 국제 표준
- Some more **Improvements** before Standardization
- 2025-2026 국가암호공모전에서 KpqC 알고리즘 관련 논문 및 고속구현 트랙 신설 (예정)



CryptoLab, Inc.





우리는 프라이버시 이슈에서 자유로운 AI, **Private AI** 를 꿈꿉니다.

- **동형암호 (FHE)**
 - 세계 최초 실수연산 동형암호 **CKKS**, 세계 최고 성능의 라이브러리 **HEaaN** 개발
 - **암호화된 상태로** 검색 (Encrypted Search), 안면인식 (Face Recognition), 사진 태그 (Photo-Tag), 검색증강생성 (RAG), 통계 연산 등 다양한 사업 진행 중
- **양자내성암호 (PQC)**
 - **미국 표준** Crystals-Kyber & Dilithium 발명가 **Damien Stehlé 교수** 수석과학자로 합류
 - 대한민국 **KpqC 표준공모 최종 선정** SMAUG-T & HAETAE 알고리즘 **개발**
 - PQC Alliance를 통해 **국내외 기업들의 양자내성암호로의 전환** 지원
 - LG U+ 광 전송 장비에 자체 양자내성암호 탑재

Thank You!